METABRIC raw omics
(mRNA, CNA, mutation, clinical)

Leakage-free stratified
train / test split
(before any fitting)

train + test
(fit on train only)

Feature aggregation
(ssGSEA pathway scores,
arm and pathway CNA burden,
recurrent mutations)

train-only
standardisation

MOFA joint
latent factorisation
(K=11 factors)

frozen
model

Choice of k by inner CV
on survival separation,
then k-means with k fixed
(survival used only for k)

out-of-fold
labels

frozen
centroids

Evaluation on
held-out patients
(C-index, AUC),
10 repeated splits,
simulation, stability,
baselines, ablations

Treatment-effect
heterogeneity
(weighted Cox,
causal forest,
E-values)

External
validation on
TCGA-BRCA,
ComBat
harmonisation

Clusters, forest plots,
figures & tables
(outputs/)